Sai Sriram Gundepalli

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EDUCATION

Eastern Illinois University — Charleston, IL
M.S., Computer Technology | GPA: 4.0 / 4.0 — May 2024

Jawaharlal Nehru Technological University — Kakinada, India
B.S., Electronics and Communication Engineering — July 2021

RESEARCH EXPERIENCE

Jawaharlal Nehru Technological University — Kakinada, India
Final Year Research Project, Dept. of Electronics & Communication — March 2021 – July 2021

IoT-Based Automated Toll Tax Collection System

- Designed an IoT-driven automated toll system integrating RFID sensors, an Arduino microcontroller, and a cloud-hosted Firebase database to automate toll transactions and vehicle verification at scale.
- Programmed Arduino modules in C/C++ for RFID tag authentication and real-time data transmission to the Firebase backend, achieving low-latency toll charge deductions.
- Developed a companion mobile application synchronized with the cloud database, enabling users to monitor trip logs, receive payment notifications, and recharge toll wallets digitally.
- Relevance: real-time data systems, distributed data processing, edge-cloud integration.

RESEARCH PROJECTS

Cluster Resource Usage Prediction & Anomaly Detection — Spring 2026

- Developed a machine learning system to model resource usage patterns in distributed computing environments using time-series data simulating real cluster workloads.
- Built regression models (Linear Regression, Random Forest) to predict CPU usage patterns and evaluated performance using standard accuracy metrics and visualization techniques.
- Implemented anomaly detection via Isolation Forest to identify abnormal system behavior, contributing to proactive resource management strategies.
- Relevance: data mining, system-level machine learning, distributed systems optimization.

PRESENTATIONS

Automatic License Number Plate Recognition Using Template Matching — May 2021

- Presented a seminar to the Department of Electronics & Communication Engineering demonstrating application of digital image processing and MATLAB GUI for automated vehicle identification.
- Explained implementation of template matching, edge detection, and morphological operations in MATLAB, achieving approximately 80% recognition accuracy for real-time vehicle license plates.
- Showcased end-to-end system design including preprocessing, segmentation, and optical character recognition (OCR), highlighting potential for traffic automation and security monitoring.
- Addressed real-world applications including car park management, traffic monitoring, tolling, and stolen vehicle detection, connecting the system to broader intelligent transportation use cases.

PROFESSIONAL EXPERIENCE

Elevance Health — Atlanta, GA
AI Engineer — May 2026 – Present

- Designed and developed a production-grade AI voice assistant backend using the Model Context Protocol (MCP), exposing a suite of callable tools to an AI voice agent over a streamable HTTP transport layer.
- Built a dual-layer LLM guardrail system (input + output) using an enterprise-internal LLM API to enforce content safety policies in real patient phone calls.
- Designed a multi-category input classifier covering several violation categories with configurable SOFT (warn/block) and STRICT (escalate to human agent) enforcement tires.

Carvintech — Atlanta, GA
Software Engineer — March 2025 – May 2026

- Designed and deployed Java 21 event-driven microservices with Kafka to process millions of customers, device, and subscription events daily, boosting throughput and cutting latency by 35% across distributed systems.
- Integrated GraphQL and REST APIs for high-volume client applications, reducing payload over-fetching by 40% and ensuring interoperability through OpenAPI/Swagger compliance.
- Engineered Postgres and Cassandra schemas with optimized indexing, partitioning, and caching strategies, achieving sub-100 ms query times for 500K+ asset and analytics queries.

SuperINTRO — San Francisco, CA
Software Engineer — December 2024 – March 2025

- Contributed to early-stage architecture of an AI-persona social platform, building event-driven Java microservices with Kafka to support real-time message delivery.
- Optimized data-streaming pipelines to achieve sub-second latency for over 10K concurrent users, improving platform responsiveness.
- Developed GraphQL APIs for structured persona data, reducing redundant API calls by 40% and enabling cross-platform client integration.

Tata Consultancy Services (TCS) — India
Software Engineer — August 2021 – November 2022

- Developed enterprise Java (Spring Boot, Hibernate) APIs for asynchronous financial workflows, cutting transaction delays by 20% across 200+ daily requests.
- Designed and optimized Oracle PL/SQL schemas and queries, enhancing retrieval speeds by 30% for large-scale data stores (500K+ records).
- Integrated REST & GraphQL APIs across departments, reducing reporting latency by 25% for 500+ workflows.

TECHNICAL SKILLS

Languages:
Java, Python, C/C++, JavaScript/TypeScript, SQL, MATLAB

AI, LLMs & Prompt Engineering:
LangChain, LangGraph, Agentic AI Architecture, MCP Agents, Guardrails, Prompt Engineering & Optimization, AI Output Validation & Hardening, AI Coding Agents (Codex, Claude, GitHub Copilot), AI-Assisted Code Generation & Refactoring, AI-Driven Workflow Automation

Frameworks & Tools:
Spring Boot, Node.js, Express, React, GraphQL, REST APIs, Kafka, Docker, Git, OpenAPI/Swagger

Databases:
PostgreSQL, Cassandra, MongoDB, Oracle PL/SQL, Redis, Firebase

Research Areas of Interest:
Distributed Systems, High-Performance Computing, Data-Intensive Systems, Machine Learning for Systems, I/O Performance Optimization

REFERENCES

Dr. David W. Melton — Eastern Illinois University
Interim Associate Dean and Interim Department Chair
dwmelton@eiu.edu

Dr. Wutthigrai Boonsuk — Eastern Illinois University
Professor and Engineering Technology Program Coordinator
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Mr. Sean T. Roberts — Eastern Illinois University
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